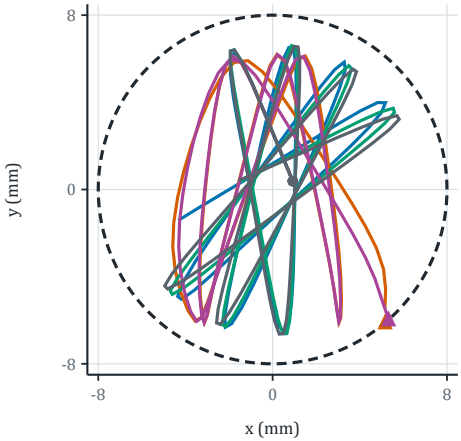
Matched dynamics reveal how a shallow angular channel becomes accessible

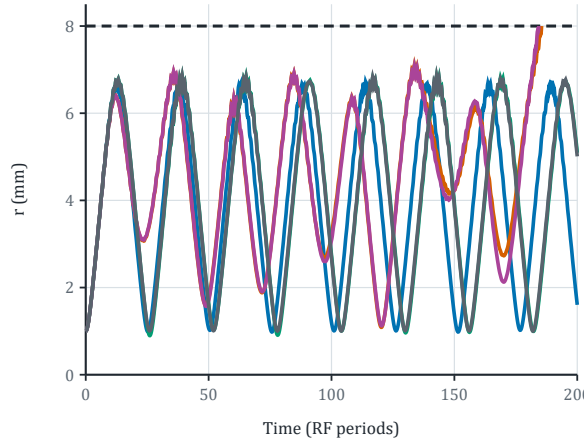
E_0 = 2 eV · r_0 = 1 mm · θ_0 = 22.5° · phase = 0° · +vθ · 250 / 353.553 V

A Stroboscopic x–y trajectories



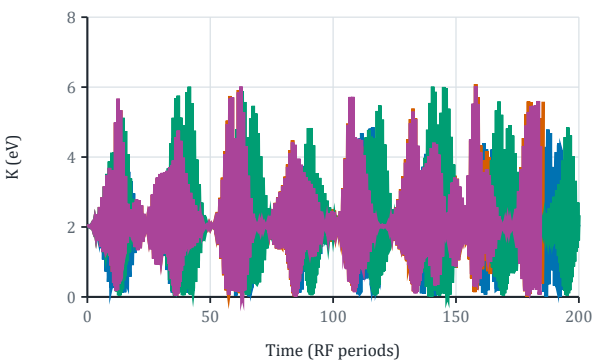
8-mm aperture

B Radius and first passage



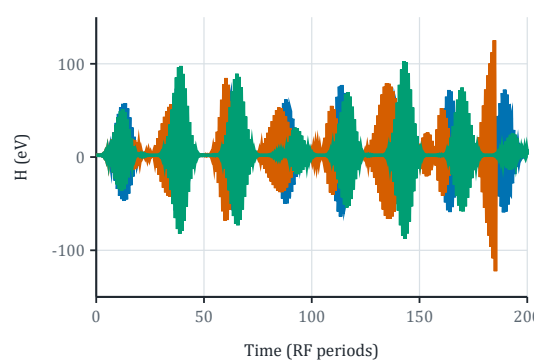
Survivors are observed to 1000 RF; plot shows first 200 RF only.

C Instantaneous kinetic energy



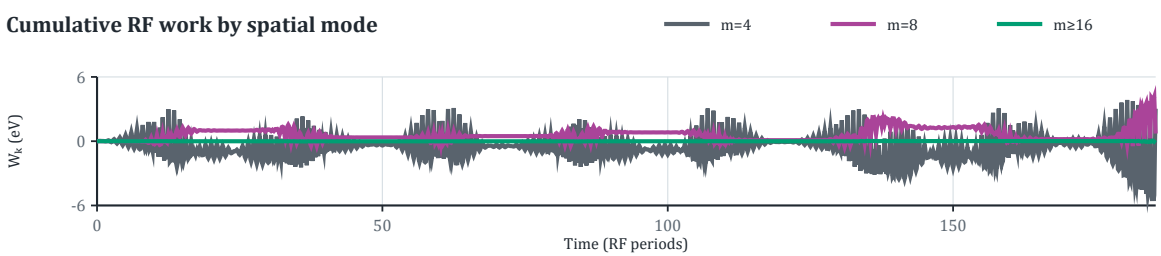
Fast micromotion remains in K(t); it is not the secular energy.

D Hamiltonian variation



Only the pure rotating m=4 control has the exact Jacobi integral.

E Cumulative RF work by spatial mode



Work is integrated as an ODE variable; $\Sigma_m W_m = \Delta K$ within 10^{-11} eV.